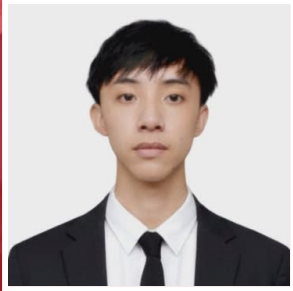




San Sebastian College - Recoletos de Cavite RESEARCH COLLOQUIUM 2026

THEME: "Pathways to Industry: Student Research and Scholarly Discovery"



“CHABALINGO: A KABITEÑO CHAVACANO WEB-BASED LANGUAGE LEARNING”

(BS Information Technology)



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Background of the Study

Technology now plays a vital role in education, especially in language learning where it enhances vocabulary, grammar, engagement, and cultural understanding. This is important for Cavite City Chavacano, a Spanish-based creole that is now spoken fluently by only a small portion of the population and is at risk of extinction. Modern tools such as mobile apps, multimedia resources, and online platforms have shown strong potential in preserving endangered languages by offering flexible learning experiences. In response, this study introduces **Chabalingo**, a web-based language platform designed to support the learning and preservation of Cavite City Chavacano, particularly among young residents. The project also supports the United Nations Sustainable Development Goals, specifically **Goal 4 (Quality Education)** and **Goal 8 (Decent Work and Economic Growth)**, by promoting cultural awareness and strengthening the city's tourism potential.



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Research Problem

1. What system requirements and design considerations are needed to develop an effective web-based language learning for the Cavite City Chavacano language?
2. How can Chabalingo be developed using appropriate web technologies and the Agile SDLC methodology to ensure usability, accessibility, and maintainability?
3. How effective is Chabalingo when evaluated in terms of system quality attributes such as functionality, reliability, performance, compatibility, security, and user satisfaction?



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Methodology

The research design used in this study is a quantitative descriptive data design, which is used in order to provide a summary of the population using numerical data. The utilization of surveys and questionnaires are used in order to measure consumer behavior and preferences, among others. Data retrieved from these surveys are then calculated and interpreted using statistical methods for the purpose of generating the mean and percentage findings, which encourages data and evidence-based decision making (Unimrkt, 2023).

Data Analysis

- Mean Formula
- Standard Deviation Formula
- User Acceptance Test (UAT) based on ISO 9126



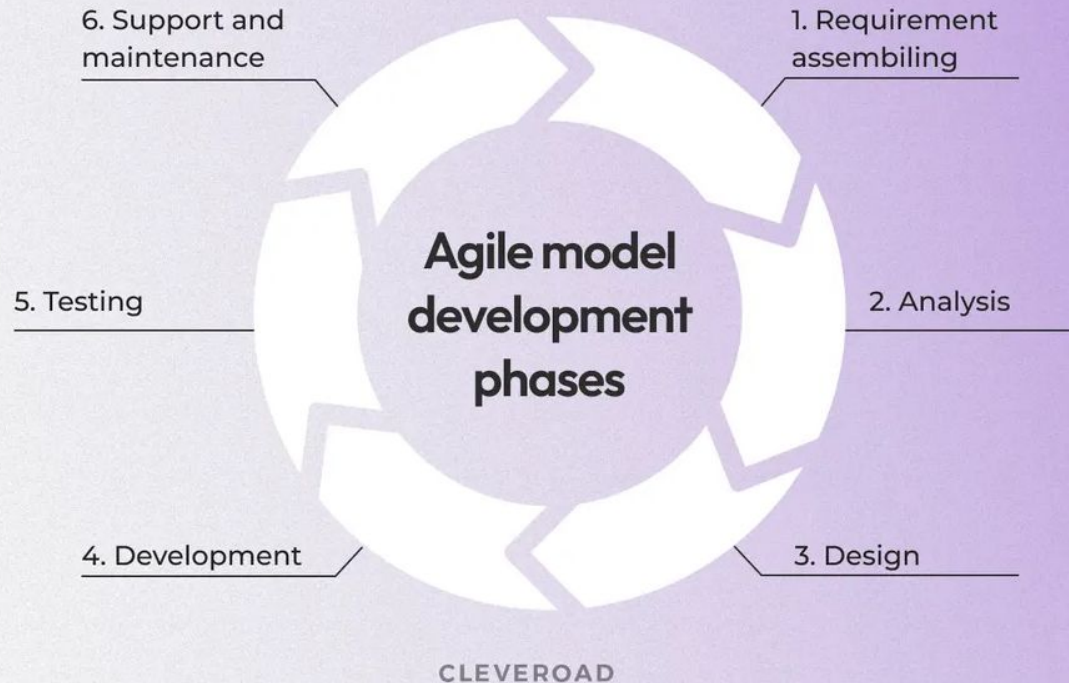
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SDLC



Data Gathering

1. Agile SDLC
2. Online Research
3. Survey Questionnaires
4. Interviews



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Results and Findings — UAT Respondents Profile

	Count	Percentage
Age Group		
Under Age	198	48.9%
18 - 24	68	16.8%
25 - 34	80	19.8%
35 - 44	41	10.1%
45 - 54	12	3%
55 and above	6	1.5%
Primary Occupation		
Student	263	64.9%
Linguist	5	1.2%
Educators	33	8.1%
IT Professional	11	2.7%
Government Employee	16	4%
Community Member	16	4%
Unemployed	44	10.9%
Retired	12	3%
Other	3	0.74%

Most of the UAT respondents are younger generation and students among 405 respondents. Aged under 18 has (48.9%), followed by 25-34 (19.8%), and 18-24 (16.8%). In terms of occupation, the majority are students (64.9%), while the unemployed individuals have (10.9%), the educators have (8.1%), and government employees and community members (4%) each.



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Results and Findings — UAT Mean & Standard Deviation

Criteria	General Responses		
	Mean	Standard Deviation	Interpolation
Functionality	3.73	0.52	Strongly Agree
Reliability	3.67	0.61	Strongly Agree
Usability	3.73	0.53	Strongly Agree
Efficiency	3.73	0.55	Strongly Agree
Maintainability	3.73	0.53	Strongly Agree
Portability	3.75	0.50	Strongly Agree
Respondent Overall	3.72	0.54	Strongly Agree

Overall Mean and Standard Deviation

Mean	Standard Deviation
3.82	0.41

Likert Scale

1	1.75	Strongly Disagree
1.76	2.50	Disagree
2.51	3.25	Agree
3.26	4	Strongly Agree



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Conclusions

This capstone study successfully designed, developed, and evaluated Chabalingo: A Kabiteño Chavacano Web-Based Language Learning. First, the study identified essential system requirements and design considerations, including user authentication with role based access control, structured course and lessons management, quiz and exam features, progress tracking, responsive design, usability, accessibility, and security. These requirements ensured that the platform met both functional and non-functional standards. Second, Chabalingo was developed by following the Agile SDLC process, which allowed the system to be built in small, manageable phases. The development started with gathering system requirements and planning the important features of the website. Next, the system was designed and developed in iterations, where features such as user accounts, courses, lessons, assessment, and admin management were implemented gradually. Each phase included testing and revisions based on feedback to improve usability and performance. Appropriate web technologies were used to create a responsive and interactive platform. Through continuous improvement and regular testing, the system achieved better usability, accessibility, and maintainability. Third, the system evaluation results showed that Chabalingo performed effectively in terms of functionality, reliability, performance, compatibility, security, and user satisfaction. The results indicate that the platform is efficient, secure, and well-received from the feedback of the users. Overall, Chabalingo is a functional and user-centered platform that supports the learning and preservation of the Cavite City Chavacano language.



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Recommendations

For the Future Researchers. Chabalingo can serve as a basis to future research in the field of educational technology platforms and language learning. They can look into how digital language learning platforms like Chabalingo actually affect language retention, keep users engaged, and help people appreciate different cultures. There's also room to dig deeper by adding features like analytics. With that, you see real performance data from learners, which gives you a better sense of how the curriculum works and how users actually learn.

For the Future Developers. In order to make learning accessible and more dynamic, future developers are encouraged to improve Chabalingo by adding additional features such as offline capability, interactive exercises, multimedia-based lessons and speech recognition that can detect what a user pronounced for comprehension evaluation. As accessibility is an important component in modern e-learning platforms it is recommended to optimize the system for mobile use.



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